

PHYS-467 Machine Learning for Physicists
Unsupervised Learning: Principal Component Analysis
November 5, 2021

Exercise 1: Eigenfaces for Facial Recognition

1. Download the ‘Labelled Faces in the Wild’ (LFW) people dataset from sklearn. This dataset is a collection of pictures of famous people faces. Set `min_faces_per_person=30`. Plot the first 10 faces (`dataset.target_images`) together with the corresponding names (`dataset.target_names`) using `imshow`.
2. In order to obtain a balanced dataset, keep only 30 pictures per person. Then, normalize your data in the range $[0, 1]$.
3. By using `sklearn.decomposition.PCA` compute the principal components and print their shape. <https://scikit-learn.org/stable/modules/generated/sklearn.decomposition.PCA.html>.
4. Plot the first five principal components, what do they look like?
5. The proportion of the variance of each principal component (i.e. eigenvector of the covariance matrix) can be computed by dividing the corresponding eigenvalue by the trace of the covariance matrix. Plot the cumulative explained variance by using the attribute `explained_variance_ratio_`. How many principal components do we need in order to explain the majority of the variance of our faces?
6. Compute and plot the projections of four pictures on the first $\{10, 30, 100, 300, 1000\}$ principal components. You can use the parameter `n_components` and the method `inverse_transform`. Comment.
7. Now, we are going to compare the performance of a naive classifier (the predicted label corresponds to the label of the nearest-neighbour image) when using the full pictures and the first 100 PCA projections. First, split your data into a training and a test set.
8. Use `sklearn.neighbors.KNeighborsClassifier` with `n_neighbors=1` to fit your training set. Evaluate the performance using the method `score` on the test set. <https://scikit-learn.org/stable/modules/generated/sklearn.neighbors.KNeighborsClassifier.html>
9. Repeat, but use only the first 100 principal components computed on the training set. Remember to project the test set also before evaluating the score. Compare with the previous score.
10. [Bonus] Before doing PCA, set the parameter `whiten=True`. This transformation, called ‘whitening’, rescales the principal components to have the same variance. Re-classify the faces and compare the accuracy with the previous ones.

Exercise 2: BBP Phase Transition in the Cards Game

1. Write a function to generate the ‘cards game dataset’. This function should generate a N -dimensional vector S with random components $S_i \in \{\pm 1\}$ and return it together with the matrix $Y = \frac{1}{\sqrt{N}}SS^T + W$, where W is a symmetric noise matrix with i.i.d. entries sampled from a Gaussian distribution $\mathcal{N}(0, \Delta)$. Be careful when generating W !
2. For $\Delta \in \{0.001, 0.01, 0.1, 1, 10\}$, diagonalize the matrix Y and plot a histogram with the spectrum. What do you observe?

3. We are going to infer the cards' values S_i using the spectral estimator $\hat{S}_i = \text{sign}(e_i^{\max})$, with $e_i^{\max}$ the i -th component of eigenvector corresponding to the largest eigenvalue. Compute the alignment

$$Q = \left| \frac{1}{N} \sum_{i=1}^N S_i \hat{S}_i \right|.$$

Notice the absolute value in order to take into account the symmetry $S \rightarrow -S$ in the problem. Compute Q for different values of Δ between 0 and 3. Average over 5 realizations for each considered value of Δ , and plot Q vs Δ . Comment. What kind of phase transition do you observe? How is the spectrum of Y related to the recovery of the two groups?